

TABLE 5—ESTIMATED EFFECT OF HAIL SCHOLARSHIP ON UNIVERSITY OF MICHIGAN APPLICATION, ADMISSION, AND ENROLLMENT BY ECONOMIC STATUS, FIRST AND SECOND HAIL COHORTS

	<i>Panel A. Free or reduced-price lunch eligibility</i>			<i>Panel B. Persistence of economic disadvantage</i>		
	Free lunch	Reduced-price	<i>p</i> -value, difference	Always disadvantaged	Sometimes disadvantaged	<i>p</i> -value, difference
Applied	0.421 (0.024) [0.270]	0.417 (0.033) [0.245]	0.921	0.422 (0.023) [0.259]	0.400 (0.043) [0.302]	0.619
Admitted	0.171 (0.022) [0.151]	0.146 (0.029) [0.146]	0.456	0.176 (0.022) [0.149]	0.151 (0.038) [0.167]	0.541
Enrolled	0.147 (0.020) [0.116]	0.115 (0.026) [0.118]	0.308	0.153 (0.020) [0.118]	0.139 (0.036) [0.130]	0.722
Number of school-years	923	607		982	425	
Number of students	2,748	1,162		3,268	642	

Notes: All analyses done at the school-year level. For each panel, treatment effects are from a single regression of school-subgroup-level outcome rate on treatment status and strata dummies, fully interacted with an indicator for subgroup. Robust standard errors clustered at the school level reported in parentheses. Control mean for subgroup in brackets. UM application, admission, and enrollment measured in the summer and fall following expected high school graduation. Admission and enrollment are unconditional on application. In Panel A, eligibility is measured in eleventh grade. In Panel B, “always disadvantaged” is defined as being eligible for free or reduced-price lunch every (observed) year of high school through 11th grade (including repeated grades).

Sources: Michigan administrative data and University of Michigan Office of Enrollment Management data

We find substantial differences in treatment effects across race and ethnic groups (see Table 6 and online Appendix Figure 8). This closely tracks the geographic variation in effects and is in fact difficult to distinguish from it. In Michigan, Black and Hispanic students are concentrated in the cities, while the rural areas are largely White. We have insufficient support in our sample to precisely estimate racial effects within region (or regional effects within race).

The intervention reduced racial and ethnic inequality in applications, admission, and enrollment at University of Michigan. Absent the intervention, students of Black, Asian, and Hispanic descent were far more likely to apply to and attend the University of Michigan than White, non-Hispanic students. For example, 21 percent of White students in the control group applied to Michigan, substantially lower than the rate for Asian (60 percent), Black (48 percent), and Hispanic students (40 percent). The pattern is similar for admission and enrollment rates.

These patterns reflect the concentration of Black, Asian, and Hispanic students in urban areas of Southeast Michigan, which are closer to the University of Michigan. The rural schools in the more distant parts of the state, which have the lowest application rates, are overwhelmingly White.

A consistent pattern in Table 6 is that the treatment effects are inversely related to control-group means. For example, HAIL’s effect on application rates is substantially larger for White students (44 percentage points) than it is for Asian, Black, and Hispanic students (23, 29, and 22 percentage points, respectively).

Finally, we examine differences by gender (see Table 6). Men are, compared to women, more confident in their skills and prefer competitive environments (Niederle